An Obstruction to the Uniform Prime Experiment in Conjecture 00000000143

quantumtool

October 19, 2026

Statement being refuted

The conjecture asks us to choose primes p_i independently and uniformly, with p_i ranging over all primes, and then makes an almost-sure claim about a random modular covering. The uniform choice required by that statement does not exist.

No uniform probability measure on the primes

The set of primes is countably infinite. Suppose a probability distribution on it assigned the same mass c to every singleton $\{p\}$. If $c > 0$, finite additivity gives mass Nc to any collection of N distinct primes. Choosing $N > 1/c$ makes this exceed the total probability 1. If $c = 0$, countable additivity gives total mass

$$\mathbb{P}(\text{a prime is chosen}) = \sum_{p \text{ prime}} \mathbb{P}(\{p\}) = 0,$$

which contradicts that the distribution has total mass 1. Hence a countably additive probability distribution that is uniform on all primes cannot exist.

More generally, on any countably infinite sample space, a probability measure must assign positive mass to at least one singleton: otherwise countable additivity makes the whole space have mass zero. Uniformity makes all singleton masses equal, so that common mass is positive, contradicting the finite-additivity bound above.

Consequence

Because the stated experiment cannot be defined, its probability-one event and the phrase "with respect to the chosen distribution" have no meaning under the claimed uniform prime law. Thus the conjecture as written is not a valid probabilistic assertion. It could be replaced by a specified non-uniform distribution on primes, but the resulting problem would be a different conjecture and could depend on that choice.

Lean arithmetic core

The accompanying self-contained Lean project formalizes the finite-additivity contradiction: for a positive common atom mass p/q, the total-mass bound would require $np \leq q$ for every finite number n of distinct atoms. Taking $n = q + 1$ is impossible. The countable-additivity step reducing the prime-space claim to this finite bound is given explicitly above.